# Solutions to Mock S(A)MC - Kevin Hu

## Question 1

**Problem:** Value of $\frac{2026^2-2025^2}{20.26+20.25}$?

*Solution.* Num: $(2026-2025)(2026+2025) = 4051$. Denom: $40.51$.
Result: $\frac{4051}{40.51} = 100$. **Ans: C** □

## Question 2

**Problem:** Smallest $n$ s.t. $2026 \mid n!$?

*Solution.* $2026 = 2 \times 1013$. 1013 is prime, so 1013! is the smallest factorial containing it. 1013! also contains 2. **Ans: B** □

## Question 3

**Problem:** $a_1 = 1, a_{n+1} = \frac{1}{1+a_n}$. Find $a_5$.

*Solution.* $a_1 = 1, a_2 = \frac{1}{2}, a_3 = \frac{2}{3}, a_4 = \frac{3}{5}, a_5 = \frac{5}{8}$. **Ans: B** □

## Question 4

**Problem:** Area ratio of Hexagon to Triangle with equal perimeters ($P = 6$).

*Solution.* We can construct a regular hexagon with sides $s = 1$ split into 6 identical unit triangles, and an equilateral triangle ($s = 2$) split into 4 unit triangles by connecting midpoints, as shown below.



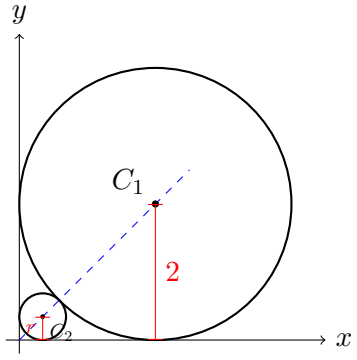
Hexagon: 6 units  Triangle: 4 units

Clearly, the area ratio $= 6 : 4 = 3/2$. **Ans: B** □

## Question 5

**Problem:** Radius $r$ of small circle tangent to large circle ($R = 2$) and axes.

*Solution.* The center of the large circle is $(2, 2)$ and the center of the small circle is $(r, r)$.






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Origin to large center distance: $2\sqrt{2} = r\sqrt{2} + r + 2$.
$r(\sqrt{2}+1) = 2\sqrt{2} - 2 \implies r = 2(\sqrt{2}-1)^2 = 6 - 4\sqrt{2}$. **Ans: A** □

## Question 6

**Problem:** Number of integers $n$ satisfying $n^4 - 20n^2 + 64 < 0$.

*Solution.* Let $x = n^2$. The inequality $x^2 - 20x + 64 < 0$ factors as $(x-4)(x-16) < 0$, giving $4 < x < 16$. Substituting back $x = n^2$, we have $4 < n^2 < 16$. The only integer $n$ whose square is strictly between 4 and 16 is $n = \pm 3$. Thus, there are 2 such integers. **Ans: A** □

## Question 7

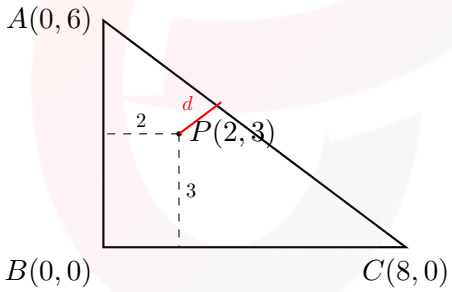
**Problem:** $N = \overline{abc}$ such that $a + b + c = 15$ and $\overline{cba} - \overline{abc} = 495$.

*Solution.* Reversal condition: $(100c + 10b + a) - (100a + 10b + c) = 99(c - a) = 495 \implies c - a = 5$. Sum of digits: $a + b + (a + 5) = 15 \implies 2a + b = 10$. Possible $(a, b, c)$ triplets: $(1, 8, 6), (2, 6, 7), (3, 4, 8), (4, 2, 9)$. Note $a = 5 \implies c = 10$ (invalid). Total 4 numbers. **Ans: E** □

## Question 8

**Problem:** $AB = 6, BC = 8, AC = 10$. $P$ is dist 2 from $AB$, 3 from $BC$. Find dist to $AC$.

*Solution.* Place $B$ at $(0, 0)$, $A$ at $(0, 6)$, and $C$ at $(8, 0)$. Then $P = (2, 3)$. Line $AC$ equation: $\frac{x}{8} + \frac{y}{6} = 1 \implies 3x + 4y - 24 = 0$.




Distance $d = \frac{|3(2)+4(3)-24|}{\sqrt{3^2+4^2}} = \frac{|6+12-24|}{5} = \frac{6}{5} = 1.2$. **Ans: D** □

## Question 9

**Problem:** Sum $S = \sum_{k=1}^{2025} f\left(\frac{k}{2026}\right)$ where $f(x) = \frac{4^x}{4^x+2}$.

*Solution.* Note $f(x) + f(1-x) = \frac{4^x}{4^x+2} + \frac{4/4^x}{4/4^x+2} = \frac{4^x}{4^x+2} + \frac{2}{2+4^x} = 1$. Pairing $f(\frac{k}{2026})$ with $f(\frac{2026-k}{2026})$ gives 1012 pairs of sum 1, plus a middle term $f(1/2) = \frac{2}{2+2} = 0.5$. Total sum $= 1012 + 0.5 = 1012.5$.
**Ans: B** □



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## Question 10

**Problem:** Probability the sum of Die A $\{1, 2, 2, 3, 3, 4\}$ and Die B $\{1, 3, 4, 5, 6, 8\}$ is 5.

*Solution.* Total outcomes: $6 \times 6 = 36$. We construct a grid of all possible sums:

| **A \ B** | **1** | **3** | **4** | **5** | **6** | **8** |
|---|---|---|---|---|---|---|
| **1** | 2 | 4 | 5 | 6 | 7 | 9 |
| **2** | 3 | 5 | 6 | 7 | 8 | 10 |
| **2** | 3 | 5 | 6 | 7 | 8 | 10 |
| **3** | 4 | 6 | 7 | 8 | 9 | 11 |
| **3** | 4 | 6 | 7 | 8 | 9 | 11 |
| **4** | 5 | 7 | 8 | 9 | 10 | 12 |

The sum of 5 occurs in **4** instances (highlighted).
Probability $= \frac{4}{36} = \frac{1}{9}$. **Ans: B** □

## Question 11

**Problem:** Sum of real $t$ for acute $A, B$ if $\sin^2 A + \cos^2 B = \frac{5}{4}t$ and $\cos^2 A + \sin^2 B = \frac{3}{4}t^2$.

*Solution.* Adding the two equations: $(\sin^2 A + \cos^2 A) + (\sin^2 B + \cos^2 B) = \frac{5}{4}t + \frac{3}{4}t^2$.
$1 + 1 = \frac{3t^2+5t}{4} \implies 3t^2 + 5t - 8 = 0$.
Factoring: $(3t + 8)(t - 1) = 0$, so $t = 1$ or $t = -8/3$.
Since $A, B$ are real, $\sin^2 A + \cos^2 B \geq 0$, so $\frac{5}{4}t \geq 0 \implies t \geq 0$.
Thus, $t = 1$ is the only valid solution. Sum $= 1$. **Ans: A** □

## Question 12 (Intended for integers not positive integers)

*Solution.* Rewrite the equation by completing the square:

$$(x + y)^2 + 2y^2 = 34$$

Since $(x + y)^2 \geq 0$, we must have $2y^2 \leq 34$, which implies $y^2 \leq 17$. The possible integer values for $y$ are $\{0, \pm1, \pm2, \pm3, \pm4\}$. We test each case for $(x + y)^2 = 34 - 2y^2$:

1. $y = 0 \implies (x + 0)^2 = 34$ (No integer solution)
2. $y = \pm1 \implies (x \pm 1)^2 = 32$ (No integer solution)
3. $y = \pm2 \implies (x \pm 2)^2 = 26$ (No integer solution)
4. $y = \pm3 \implies (x \pm 3)^2 = 34 - 18 = 16 \implies x \pm 3 = \pm4$
   - If $y = 3$: $x + 3 = 4 \implies x = 1$; or $x + 3 = -4 \implies x = -7$.
   - If $y = -3$: $x - 3 = 4 \implies x = 7$; or $x - 3 = -4 \implies x = -1$.
5. $y = \pm4 \implies (x \pm 4)^2 = 34 - 32 = 2 \implies$ (No integer solution)

The complete set of integer solutions $(x, y)$ is:

$$\{(1, 3), (-7, 3), (7, -3), (-1, -3)\}$$

There are **4** such pairs. **Ans: D** □



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## Question 13

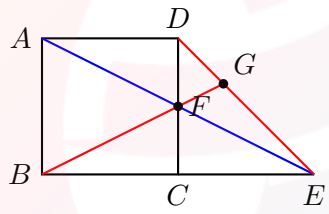
**Problem:** Min value of $a^4 + ab + b^4$ given $a^2 + b^2 = 1$.

*Solution.* $a^4 + b^4 = (a^2 + b^2)^2 - 2a^2b^2 = 1 - 2(ab)^2$.
Let $k = ab$. Since $|ab| \leq \frac{a^2+b^2}{2}$, we have $-\frac{1}{2} \leq k \leq \frac{1}{2}$.
Expression $f(k) = 1 - 2k^2 + k$. This is a downward parabola with vertex at $k = \frac{-1}{2(-2)} = 1/4$.
Max occurs at $k = 1/4$, but we want the **minimum**. Check endpoints:
$f(1/2) = 1 - 2(1/4) + 1/2 = 1$.
$f(-1/2) = 1 - 2(1/4) - 1/2 = 0$.
The minimum value is 0. **Ans: A** □

## Question 14

**Problem:** $ABCD$ unit square, $CE = 1$. $AE \cap DC = F$, $BF \cap DE = G$. Find $BG$.

*Solution.* We can appraoch using simple coordinate geometry bash. Place $B$ at $(0, 0)$, $A$ at $(0, 1)$, $C$ at $(1, 0)$, and $D$ at $(1, 1)$. Since $CE = 1$ and $E$ is on $BC$ extended, $E = (2, 0)$.
1. Find $F$: Line $AE$ passes through $(0, 1)$ and $(2, 0) \implies y = -\frac{1}{2}x + 1$. Since $F$ is on $CD$ ($x = 1$), $y_F = -\frac{1}{2}(1) + 1 = 0.5$. So $F = (1, 0.5)$.
2. Find $G$: Line $BF$ passes through $(0, 0)$ and $(1, 0.5) \implies y = 0.5x$. Line $DE$ passes through $(1, 1)$ and $(2, 0) \implies y = -x + 2$. Equating them: $0.5x = -x + 2 \implies 1.5x = 2 \implies x = 4/3, y = 2/3$. Thus $G = (4/3, 2/3)$.




3. Calculate $BG$: $BG = \sqrt{(4/3)^2 + (2/3)^2} = \sqrt{16/9 + 4/9} = \sqrt{20/9} = \frac{2\sqrt{5}}{3}$. **Ans: D** □

## Question 15

**Problem:** Min value of $f(x) = \sum_{k=0}^{2017} |x - k|$.

*Solution.* The sum of absolute differences is minimized at the median of the data points.
Points are $\{0, 1, \ldots, 2017\}$ (2018 points). Median is any value in $[1008, 1009]$.
Let $x = 1008$. $f(1008) = \sum_{k=0}^{2017} |1008 - k| = 2\sum_{j=1}^{1008} j + 1009 = 2\frac{1008 \times 1009}{2} + 1009$.
$1008 \times 1009 + 1009 = 1009^2 = 1018081$. **Ans: B** □

## Question 16

**Problem Statement:** Let $S = \sum_{n=1}^{2017} \frac{1}{n^3}$. Find the value of $\lfloor 4S \rfloor$.

*Solution.* To determine $\lfloor 4S \rfloor$, we approximate $S$ by calculating the initial terms and bounding the remaining "tail" of the series.



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**1. Grouping**

We separate the first three terms and group the remaining terms to create an upper bound:

$$S = 1 + \frac{1}{8} + \frac{1}{27} + \sum_{n=4}^{2017} \frac{1}{n^3}$$

$$S < 1 + \underbrace{\left(\frac{1}{2^3} + \frac{1}{3^3}\right)}_{\text{Group 1}} + \underbrace{\left(\frac{1}{4^3} + \cdots + \frac{1}{7^3}\right)}_{\text{Group 2}} + \underbrace{\left(\frac{1}{8^3} + \cdots + \frac{1}{15^3}\right)}_{\text{Group 3}} + \dots$$

**2. Hidden geometric series**

By comparing the sum to a geometric series where each block of $2^k$ terms is bounded by the first term of the block, we get:

$$S < 1 + \frac{1}{8} + \frac{1}{27} + \left[\frac{1}{16} + \frac{1}{64} + \frac{1}{256} + \dots\right]$$

$$S < 1.1620 + \sum_{k=2}^{\infty} \left(\frac{1}{4}\right)^k$$

The sum of the infinite geometric series $\sum_{k=2}^{\infty}(\frac{1}{4})^k$ with first term $a = \frac{1}{16}$ and ratio $r = \frac{1}{4}$ is:

$$\text{Tail} = \frac{1/16}{1 - 1/4} = \frac{1/16}{3/4} = \frac{1}{12} \approx 0.0833$$

**3. Approximation**

Adding the components together:

$$S < 1.1620 + 0.0833 = 1.2453$$

Multiplying by 4 to find the bounds for $4S$:

- **Upper Bound:** $4S < 4 \times 1.2453 = 4.9812$
- **Lower Bound:** Since $S > 1 + \frac{1}{8} + \frac{1}{27} \approx 1.1620$, then $4S > 4 \times 1.1620 = 4.648$

Since $4.648 < 4S < 4.9812$, the integer part must be 4. One can also add up $1 + \frac{1}{8} + \frac{1}{27} + \frac{1}{12}$, and then multiply by 4, which is fairly trivial to compute.

**Ans: B** □

## Question 17

**Problem:** Find the mean of all values of $x + \frac{1}{x}$ for $4x^4 + 3x^3 - 2x^2 + 3x + 4 = 0$.

*Solution.* This is a reciprocal equation. Divide by $x^2$:

$$4x^2 + 3x - 2 + \frac{3}{x} + \frac{4}{x^2} = 0 \implies 4\left(x^2 + \frac{1}{x^2}\right) + 3\left(x + \frac{1}{x}\right) - 2 = 0$$

Let $u = x + \frac{1}{x}$. Then $x^2 + \frac{1}{x^2} = u^2 - 2$. Substitute:

$$4(u^2 - 2) + 3u - 2 = 0 \implies 4u^2 + 3u - 10 = 0$$

This is a quadratic in $u$. The roots $u_1, u_2$ represent the possible values of $x + \frac{1}{x}$. By Vieta's formulas, the sum of the roots is $u_1 + u_2 = -b/a = -3/4$. The mean is $\frac{u_1+u_2}{2} = \frac{-3/4}{2} = -3/8$. **Ans: B** □



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## Question 18

**Problem:** If $x + \frac{1}{x} = \sqrt{5}$, what is $x^{11} + \frac{1}{x^{11}}$?

*Solution.* Let $V_n = x^n + x^{-n}$. We use the recurrence $V_n = V_1 V_{n-1} - V_{n-2}$.

$$V_0 = 2 \longrightarrow V_1 = \sqrt{5} \longrightarrow V_2 = 3 \longrightarrow V_3 = 2\sqrt{5}$$
$$\dots V_n = \sqrt{5} V_{n-1} - V_{n-2} \dots$$

Iterating: $V_4 = 7$, $V_5 = 5\sqrt{5}$, $V_6 = 18$. Using the identity $V_{m+n} = V_m V_n - V_{m-n}$:

$$V_{11} = V_6 V_5 - V_1 = (18)(5\sqrt{5}) - \sqrt{5} = 90\sqrt{5} - \sqrt{5} = 89\sqrt{5}$$

**Ans: C** □

## Question 19

**Problem:** $P(k) = \frac{k}{k+1}$ for $k = 0, \dots, n$. Find $P(n+1)$.

*Solution.* Define $Q(x) = (x+1)P(x) - x$. $Q(x)$ is a polynomial of degree $n+1$ with roots at $x = 0, 1, \dots, n$. Thus, $Q(x) = C \cdot x(x-1)(x-2)\dots(x-n)$. To find $C$, let $x = -1$:

$$Q(-1) = 0 - (-1) = 1 \implies 1 = C(-1)(-2)\dots(-(n+1)) = C(-1)^{n+1}(n+1)!$$

So $C = \frac{(-1)^{n+1}}{(n+1)!}$. Now evaluate $Q(n+1)$:

$$(n+2)P(n+1) - (n+1) = \frac{(-1)^{n+1}}{(n+1)!}(n+1)! = (-1)^{n+1}$$

Solving for $P(n+1)$:

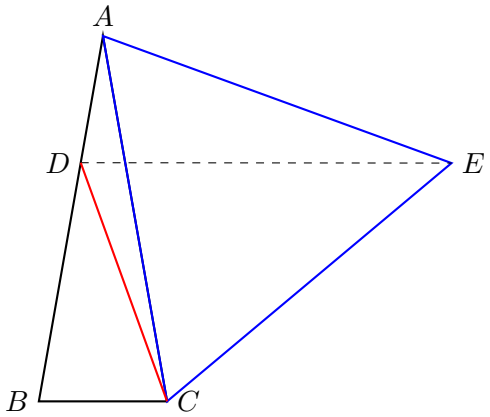
$$P(n+1) = \frac{n+1+(-1)^{n+1}}{n+2}$$

**Ans: C** □

## Question 20

**Problem:** In isosceles $\triangle ABC$, $\angle A = 20^\circ$ and $AB = AC$. $D$ lies on $AB$ such that $AD = BC$. Find $\angle BDC$.

*Solution 1 - pure geometry.* In $\triangle ABC$, $\angle B = \angle C = 80^\circ$. Construct equilateral $\triangle ACE$ externally. Connect $DE$ and $DC$.






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1. In $\triangle ADE$ and $\triangle CAB$: $AD = BC$ (given), $AE = AC$ (equilateral), and $\angle EAD = 60° + 20° = 80°$. By SAS, $\triangle ADE \cong \triangle CAB$.

2. From congruence, $ED = AB$. Since $AB = AC = EC$, $\triangle DEC$ is isosceles with $ED = EC$.

3. - $\angle ADE = \angle ABC = 80°$ and $\angle AED = \angle BAC = 20°$.
   - $\angle DEC = \angle AEC - \angle AED = 60° - 20° = 40°$.
   - Thus, the base angle $\angle EDC = (180° - 40°)/2 = 70°$.

4. $\angle BDC = 180° - \angle ADE - \angle EDC = 180° - 80° - 70° = 30°$.

**Ans: C** □

*Solution 2 - Trig bash (Not Recommended).* Normalize by letting $BC = AD = 1$. In $\triangle ABC$, the Sine Rule gives:

$$\frac{1}{\sin 20°} = \frac{AB}{\sin 80°} \implies AB = \frac{\sin 80°}{\sin 20°}$$

In $\triangle BDC$, we have $\angle B = 80°$ and $BD = AB - 1$. By the Sine Rule:

$$\frac{BD}{\sin(100° - x)} = \frac{1}{\sin x} \implies \sin x(AB - 1) = \sin(100° - x)$$

Substitute $AB$:

$$\sin x \left( \frac{\sin 80° - \sin 20°}{\sin 20°} \right) = \sin(100° - x)$$

Using $\sin 80° - \sin 20° = 2 \cos 50° \sin 30° = \cos 50° = \sin 40°$:

$$\sin x \left( \frac{\sin 40°}{\sin 20°} \right) = \sin x(2 \cos 20°) = \sin(100° - x)$$

Expanding the right side:

$$2 \sin x \cos 20° = \sin 100° \cos x - \cos 100° \sin x$$

Divide by $\sin x$ and use $\sin 100° = \cos 10°, \cos 100° = -\sin 10°$:

$$2 \cos 20° = \cos 10° \cot x + \sin 10°$$

$$\cot x = \frac{2 \cos 20° - \sin 10°}{\cos 10°} = \frac{\sqrt{3} \cos 10° + \sin 10° - \sin 10°}{\cos 10°} = \sqrt{3}$$

Thus, $x = 30°$. **Ans: C** □

## Question 21

**Problem:** If $a, b, c$ are roots of $x^3 - 5x^2 + 7x - 9 = 0$, find $S = \frac{a^2}{bc} + \frac{b^2}{ac} + \frac{c^2}{ab}$.

*Solution.* The expression simplifies to $S = \frac{a^3 + b^3 + c^3}{abc}$. From Vieta's formulas:

- $e_1 = a + b + c = 5$
- $e_2 = ab + bc + ca = 7$
- $e_3 = abc = 9$



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Using the identity $a^3+b^3+c^3-3abc=(a+b+c)(a^2+b^2+c^2-(ab+bc+ca))$:

1. $a^2+b^2+c^2=e_1^2-2e_2=5^2-2(7)=11$.
2. $a^3+b^3+c^3-3(9)=5(11-7) \implies a^3+b^3+c^3-27=20$.
3. $a^3+b^3+c^3=47$.

Thus, $S=\frac{47}{9}$. **Ans: B** □

## Question 22

**Problem:** The number $N=\underbrace{11\ldots1}_{81}$ is divisible by which power of 3?

*Solution.* A repunit $R_n$ is divisible by $3^k$ if $n$ is divisible by $3^{k-1}$ (for $n>1$). Formally, using the Lifting The Exponent Lemma for $p=3$:

$$v_3(R_n)=v_3\left(\frac{10^n-1}{9}\right)=v_3(10^n-1)-v_3(9)$$

Since $v_3(10-1)=2$, for $n=81$:

$$v_3(10^{81}-1)=v_3(10-1)+v_3(81)=2+4=6$$

Then $v_3(N)=6-2=4$. So $N$ is divisible by $3^4=81$. **Ans: A** □

## Question 23

**Problem:** Sum of $p$ for $x^2+2px-3p-2=0$ if $(x_1^3+x_2^3)+(x_1^2+x_2^2)=4$.

*Solution.* Vieta's: $s=x_1+x_2=-2p$ and $q=x_1x_2=-3p-2$.

- $x_1^2+x_2^2=s^2-2q=4p^2+6p+4$.
- $x_1^3+x_2^3=s(s^2-3q)=-2p(4p^2+9p+6)=-8p^3-18p^2-12p$.

Equation: $(-8p^3-18p^2-12p)+(4p^2+6p+4)=4 \implies -8p^3-14p^2-6p=0$. Factoring: $-2p(4p+3)(p+1)=0$. The values are $p=0,p=-1,p=-3/4$. The discriminant $\Delta=4(p+1)(p+2)$ must be $>0$ for real distinct roots. This excludes $p=-1$. Remaining $p\in\{0,-3/4\}$. Sum $=-3/4$. **Ans: B** □

## Question 24

**Problem:** $abc=-1, a+b+c=4, \sum\frac{a}{a^2-3a-1}=4/9$. Find $a^2+b^2+c^2$.

*Solution.* Since $abc=-1$, the denominator $a^2-3a-1$ can be written as $a^2-3a+abc=a(a-3+bc)$. As $a-3=1-b-c$, the expression becomes $a(bc-b-c+1)=a(b-1)(c-1)$. The sum is:

$$\sum_{cyc}\frac{1}{(b-1)(c-1)}=\frac{(a-1)+(b-1)+(c-1)}{(a-1)(b-1)(c-1)}=\frac{(a+b+c)-3}{abc-(ab+bc+ca)+(a+b+c)-1}$$

Let $x=ab+bc+ca$. Then:

$$\frac{4-3}{-1-x+4-1}=\frac{1}{2-x}=\frac{4}{9} \implies x=-1/4$$

$a^2+b^2+c^2=x^2-2x=16-2(-1/4)=\frac{33}{2}$. **Ans: E** □



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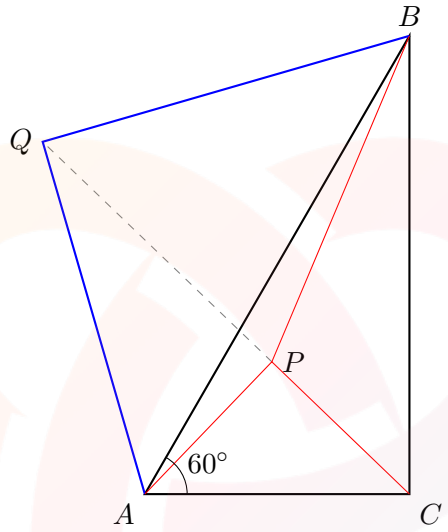
## Question 25

**Problem Statement:** In $\triangle ABC$, $\angle A = 60°$ and $AB = 2AC$. Let $P$ be a point **inside** the triangle such that $PA = \sqrt{3}$, $PB = 5$, and $PC = 2$. Find the area of $\triangle ABC$.

*Solution.* Since $AB = 2AC$ and $\angle A = 60°$, $\triangle ABC$ is a 30°-60°-90° triangle with $\angle ACB = 90°$ and $\angle ABC = 30°$.

**1. Construction**

Construct $\triangle ABQ$ outside $\triangle ABC$ such that $\triangle ABQ \sim \triangle ACP$ via a spiral similarity centered at $A$. Specifically, $\angle QAB = \angle PAC$ and $\frac{AQ}{AB} = \frac{AP}{AC}$. Since $AB/AC = 2$, it follows that $AQ = 2AP = 2\sqrt{3}$ and $BQ = 2CP = 4$.




**2. Intermediate Relationships**

The angle $\angle QAP = \angle QAB + \angle BAP = \angle PAC + \angle BAP = \angle BAC = 60°$.
In $\triangle APQ$, since $AQ = 2AP$ and $\angle QAP = 60°$, it is a 30°-60°-90° triangle. Thus:

- $PQ = \sqrt{3}AP = \sqrt{3}(\sqrt{3}) = 3$.
- $\angle APQ = 90°$.

**3. Cosine Rule**

In $\triangle BQP$, the sides are $BQ = 4$, $PQ = 3$, and $PB = 5$. Since $3^2 + 4^2 = 5^2$, $\angle BQP = 90°$ (DIAGRAM NOT TO SCALE) by the Pythagorean converse.
Now, in $\triangle ABQ$, $\angle AQB = \angle AQP + \angle BQP = 30° + 90° = 120°$.
By the Law of Cosines in $\triangle ABQ$:

$$AB^2 = AQ^2 + BQ^2 - 2(AQ)(BQ)\cos(120°)$$
$$AB^2 = (2\sqrt{3})^2 + 4^2 - 2(2\sqrt{3})(4)(-1/2)$$
$$AB^2 = 12 + 16 + 8\sqrt{3} = 28 + 8\sqrt{3}$$

**4. Area**

For a 30°-60°-90° triangle with hypotenuse $AB$:

$$\text{Area} = \frac{1}{2} \cdot \left(\frac{AB}{2}\right) \cdot \left(\frac{AB\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{8} AB^2$$



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$$\text{Area} = \frac{\sqrt{3}}{8}(28 + 8\sqrt{3}) = \frac{28\sqrt{3} + 24}{8} = \frac{7\sqrt{3} + 6}{2}$$

**Ans: E** □



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